Sum and Difference Formulas

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.03

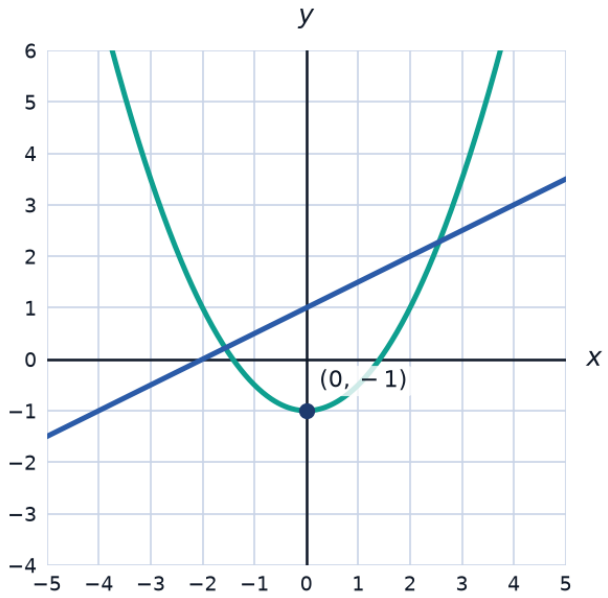
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Exact values with sum and difference formulas

Use a sum or difference formula to find the exact value of $\cos 75°$.

2 [Foundational] • Target: Exact values with sum and difference formulas

Find the exact value of $\sin \frac{\pi}{12}$.



3 [Foundational] • Target: Exact values with sum and difference formulas

Find the exact value of $\tan 105°$.

4 [Foundational] • Target: Working with given values

Angles A and B are both in Quadrant I with $\sin A = \frac{4}{5}$ and $\cos B = \frac{5}{13}$. Find:

a $\sin(A + B)$

b $\cos(A + B)$

c the quadrant of $A + B$

5 [Foundational] • Target: Working with given values

Use a sum or difference formula to simplify:

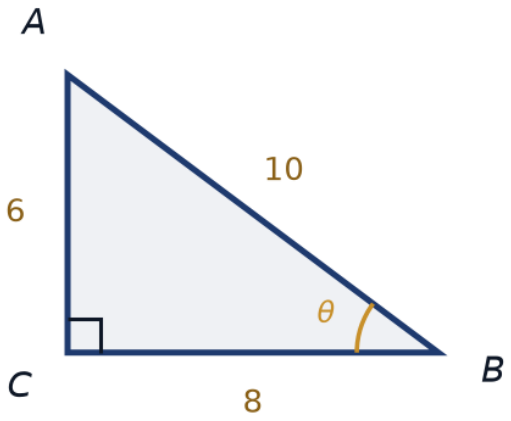
a $\sin\left(x + \frac{\pi}{2}\right)$

b $\cos(x - \pi)$

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6 [Foundational] • Target: Working with given values

Angles A and B are both in Quadrant II with $\sin A = \frac{8}{17}$ and $\sin B = \frac{3}{5}$. Find $\cos(A - B)$ exactly.



7 [Foundational] • Target: Verifying identities using sum and difference formulas

Verify the identity $\cos(x + y) + \cos(x - y) = 2\cos x \cos y$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Verifying identities using sum and difference formulas

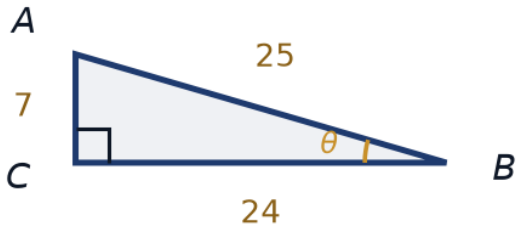
Verify the identity $\sin(x + y) - \sin(x - y) = 2\cos x \sin y$.

9 [Intermediate] • Target: Verifying identities using sum and difference formulas

Show that $\tan\left(x + \frac{\pi}{4}\right) = \frac{1 + \tan x}{1 - \tan x}$ using the tangent sum formula $\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$.

10 [Intermediate] • Target: Applications

Two musical tones combine to form a sound wave modeled by $y = \sin\left(x + \frac{\pi}{6}\right)$. Expand this expression using the sum formula, and evaluate the coefficient of $\cos x$ exactly.



11 [Intermediate] • Target: Additional practice

Find the exact value of $\cos\frac{\pi}{12}$.

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12 [Intermediate] • Target: Additional practice

Angles A and B satisfy $\tan A = \frac{3}{4}$ (QI) and $\tan B = \frac{5}{12}$ (QI). Find $\tan(A + B)$ using the tangent sum formula.

13 [Intermediate] • Target: Additional practice

Simplify $\cos\left(\frac{\pi}{2} - x\right)\cos x + \sin\left(\frac{\pi}{2} - x\right)\sin x$ using the cosine difference formula in reverse.

14 [Intermediate] • Target: A worked derivation

Starting from the cosine difference formula $\cos(A - B) = \cos A\cos B + \sin A\sin B$, derive the cosine sum formula $\cos(A + B)$ by substituting $B \to -B$.

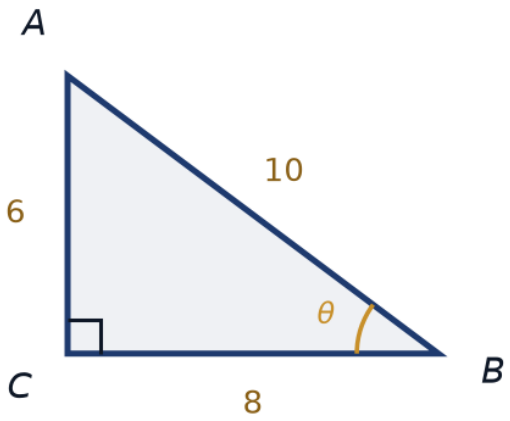
Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: A worked derivation • Cross-Lesson: 8.02 Graphing Rational Functions

Find the exact value of $\sin 15°\cos 15°$ using a sum/difference-related double angle shortcut (recall $\sin(2\theta) = 2\sin\theta\cos\theta$).

16 [Challenging] • Target: Sum and difference formulas with tables of values • Cross-Lesson: 8.03 Introduction to Limits

Given $\sin A = \frac{7}{25}$ (QI) and $\cos B = \frac{4}{5}$ (QI), find $\sin(A - B)$ exactly.



17 [Challenging] • Target: Sum and difference formulas with tables of values • Cross-Lesson: 9.01 Arithmetic Sequences and Series

Given $\tan A = 2$ (QI) and $\tan B = 3$ (QI), find $\tan(A - B)$ using the tangent difference formula.

18 [Challenging] • Target: Expanding a general sinusoidal expression • Cross-Lesson: 9.02 Geometric Sequences and Series

Expand $\cos\left(x - \frac{\pi}{3}\right)$ fully using the difference formula, leaving the answer in terms of $\sin x$ and $\cos x$.

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19 [Challenging] • Target: A geometric interpretation • Cross-Lesson: 9.03 The Binomial Theorem

Explain, using the unit circle, why $\cos(A - B)$ (not $\cos A - \cos B$) correctly gives the cosine of the angle between two points at angles A and B on the unit circle.

20 [Challenging] • Target: Solving equations using sum and difference formulas • Cross-Lesson: 1.01 Functions and Function Notation

Solve $\sin\left(x + \frac{\pi}{6}\right) = \frac{1}{2}$ for $x \in [0, 2\pi)$ by first expanding the left side.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Sine Sum and Difference Identities

State the formula for $\sin(A + B)$.

a. $\sin A \cos B + \cos A \sin B$

b. $\sin A \cos B - \cos A \sin B$

c. $\cos A \cos B - \sin A \sin B$

d. $\sin A + \sin B$

22 [Foundational] • Target: Sine Sum and Difference Identities

Find the exact value of $\sin(75^\circ)$ using $75^\circ = 45^\circ + 30^\circ$.

a. $\frac{\sqrt{6} + \sqrt{2}}{4}$

b. $\frac{\sqrt{6} - \sqrt{2}}{4}$

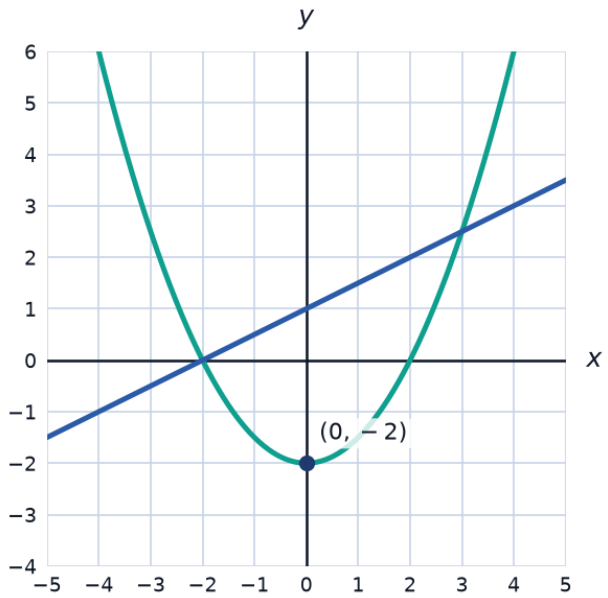
c. $\frac{\sqrt{3} + 1}{2}$

d. $\frac{\sqrt{2} + 1}{2}$

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23 [Foundational] • Target: Sine Sum and Difference Identities

Simplify $\sin(x + \pi)$.



a. $-\sin x$

b. $\sin x$

c. $-\cos x$

d. $\cos x$

24 [Foundational] • Target: Cosine Sum and Difference Identities

State the formula for $\cos(A - B)$.

a. $\cos A \cos B + \sin A \sin B$

b. $\cos A \cos B - \sin A \sin B$

c. $\sin A \sin B - \cos A \cos B$

d. $\cos A - \cos B$

25 [Foundational] • Target: Cosine Sum and Difference Identities

Find the exact value of $\cos(15^\circ)$ using $15^\circ = 45^\circ - 30^\circ$.

a. $\frac{\sqrt{6}+\sqrt{2}}{4}$

b. $\frac{\sqrt{6}-\sqrt{2}}{4}$

c. $\frac{\sqrt{3}-1}{4}$

d. $\frac{\sqrt{2}+1}{4}$

26 [Foundational] • Target: Cosine Sum and Difference Identities

Simplify $\cos(x - \pi/2)$.

a. $\sin x$

b. $-\sin x$

c. $\cos x$

d. $-\cos x$

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27 [Foundational] • Target: Tangent Sum and Difference Identities

State the formula for $\tan(A+B)$.

a. $\frac{\tan A+\tan B}{1-\tan A\tan B}$

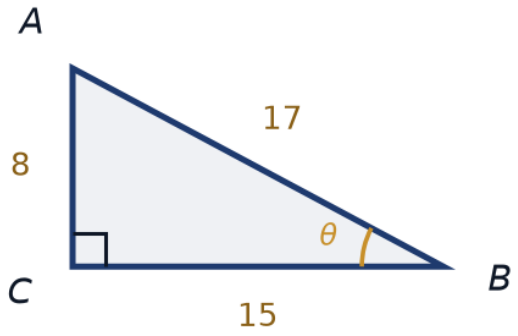
b. $\frac{\tan A-\tan B}{1+\tan A\tan B}$

c. $\frac{\tan A+\tan B}{1+\tan A\tan B}$

d. $\tan A+\tan B$

28 [Foundational] • Target: Tangent Sum and Difference Identities

If $\tan A=1/2$ and $\tan B=1/3$, find $\tan(A+B)$.



a. 1

b. 5/6

c. 5/7

d. 2/3

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Tangent Sum and Difference Identities

Evaluate $\frac{\tan(80^\circ)-\tan(20^\circ)}{1+\tan(80^\circ)\tan(20^\circ)}$.

a. $\sqrt{3}$

b. $1/\sqrt{3}$

c. 1

d. 0

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30 [Intermediate] • Target: Compound Angle Problems with Given Ratios

Given $\sin\alpha = 3/5$ (Quadrant I) and $\cos\beta = -5/13$ (Quadrant II), find $\sin(\alpha+\beta)$.

a. $33/65$

b. $-33/65$

c. $63/65$

d. $-63/65$

31 [Intermediate] • Target: Compound Angle Problems with Given Ratios

For the same angles ($\cos\alpha = 4/5, \sin\alpha = 3/5, \cos\beta = -5/13, \sin\beta = 12/13$), find $\cos(\alpha+\beta)$.

a. $-56/65$

b. $56/65$

c. $-16/65$

d. $16/65$

32 [Intermediate] • Target: Compound Angle Problems with Given Ratios

In which quadrant does $\alpha+\beta$ lie in the previous problems?

a. Quadrant II

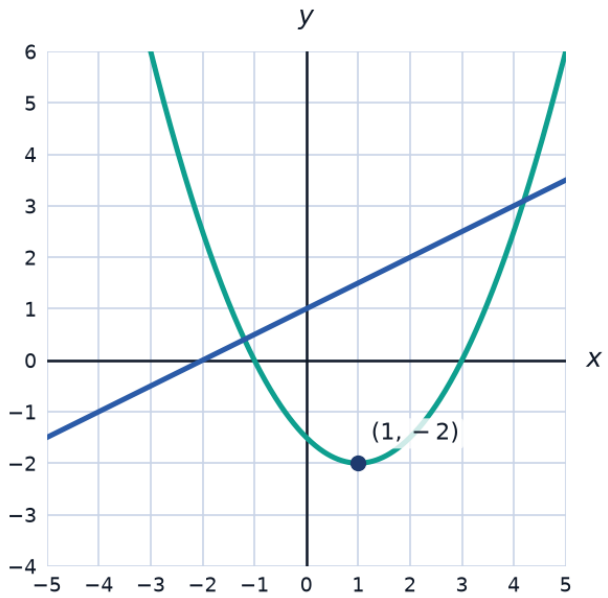
b. Quadrant I

c. Quadrant III

d. Quadrant IV

33 [Intermediate] • Target: Tangent Sum and Difference Identities

Evaluate $\sin(45^\circ + 30^\circ)$ using a sum/difference formula. (variant 10)



a. $6-24$

b. 22

c. $2-64$

d. $6+24$

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34 [Intermediate] • Target: Compound Angle Problems with Given Ratios

Evaluate $\cos(45^\circ - 30^\circ)$ using a sum/difference formula. (variant 5)

a. $6 + 24$

b. $2 - 64$

c. 32

d. $6 - 24$

35 [Intermediate] • Target: Exact values with sum and difference formulas

Evaluate $\cos(60^\circ + 45^\circ)$ using a sum/difference formula. (variant 12)

a. $6 + 24$

b. $6 - 24$

c. $\frac{1}{2}$

d. $2 - 64$

36 [Intermediate] • Target: Working with given values

Evaluate $\cos(45^\circ - 30^\circ)$ using a sum/difference formula. (variant 10)

a. $2 - 64$

b. 32

c. $\frac{1}{2}$

d. $6 + 24$

37 [Intermediate] • Target: Verifying identities using sum and difference formulas

Evaluate $\sin(60^\circ - 45^\circ)$ using a sum/difference formula. (variant 7)

a. $6 + 24$

b. 22

c. $6 - 24$

d. $\frac{1}{2}$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Applications • Cross-Lesson: 9.02 Geometric Sequences and Series

Evaluate $\sin(60^\circ - 45^\circ)$ using a sum/difference formula. (variant 9)

a. $2 - 64$

b. 22

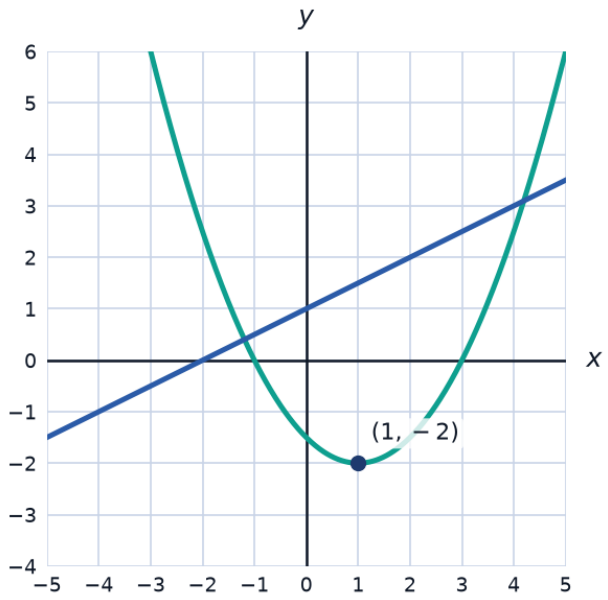
c. $6 + 24$

d. $6 - 24$

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39 [Challenging] • Target: Additional practice • Cross-Lesson: 9.03 The Binomial Theorem

Evaluate $\cos(45^\circ - 30^\circ)$ using a sum/difference formula. (variant 7)



a. $6 + 24$

b. $2 - 64$

c. 32

d. 22

40 [Challenging] • Target: A worked derivation • Cross-Lesson: 1.01 Functions and Function Notation

Evaluate $\sin(45^\circ + 30^\circ)$ using a sum/difference formula. (variant 3)

a. $2 - 64$

b. 32

c. 22

d. $6 + 24$

41 [Challenging] • Target: Sum and difference formulas with tables of values • Cross-Lesson: 1.02 Domain and Range

Evaluate $\cos(60^\circ + 45^\circ)$ using a sum/difference formula.

a. $2 - 64$

b. $\frac{1}{2}$

c. 22

d. $6 - 24$

42 [Challenging] • Target: Expanding a general sinusoidal expression • Cross-Lesson: 1.03 Average Rate of Change

Evaluate $\sin(45^\circ + 30^\circ)$ using a sum/difference formula. (variant 8)

a. $2 - 64$

b. $\frac{1}{2}$

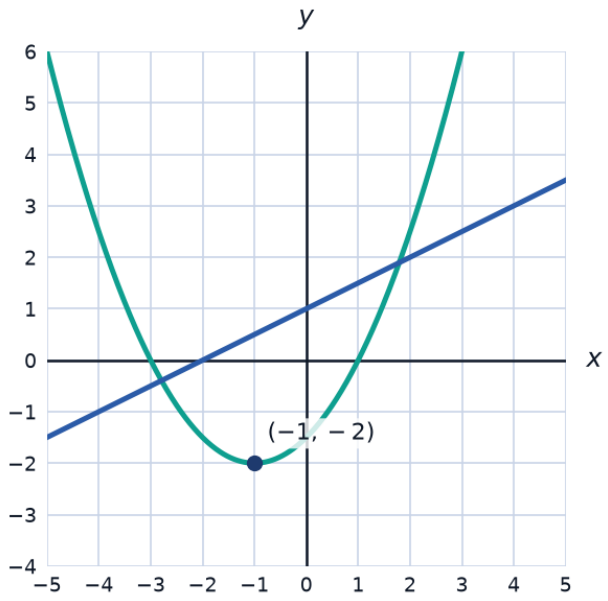
c. $6 + 24$

d. $6 - 24$

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43 [Challenging] • Target: A geometric interpretation • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Evaluate $\sin(45^\circ + 30^\circ)$ using a sum/difference formula. (variant 2)



a. $\frac{1}{2}$

b. $6 + 24$

c. $2 - 64$

d. 22

44 [Challenging] • Target: Solving equations using sum and difference formulas • Cross-Lesson: 2.02 Radian Measure and Arc Length

Evaluate $\sin(60^\circ - 45^\circ)$ using a sum/difference formula. (variant 3)

a. 32

b. $\frac{1}{2}$

c. $6 + 24$

d. $6 - 24$

45 [Challenging] • Target: Sine Sum and Difference Identities • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Evaluate $\cos(60^\circ + 45^\circ)$ using a sum/difference formula. (variant 4)

a. $6 + 24$

b. $2 - 64$

c. $6 - 24$

d. 32

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